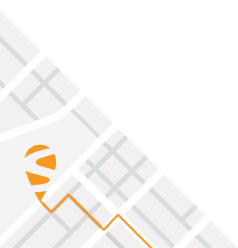




Release Notes and Installation Guide

mosaic-G5 P3 Firmware Package v1.0.0



1 Installation Guidelines

To upgrade the firmware to version 1.0.0, upgrade the receiver with the suf file `mosaic-G5 P3-1.0.0.suf` as explained in the section **Upgrade the Receiver** of the mosaic-G5 P3 Reference Guide.

Upgrading the receiver's failsafe with `mosaic-G5 P3-1.0.0-failsafe.suf` is not necessary, unless Septentrio Support advises otherwise.

2 New Features and Improvements

2.1 New features in version 1.0.0

1. The command `setSolutionSelectivity` was added for users to tune RTK availability depending on its application environment.
2. The command `setSatelliteHealthOverride` and `setSignalHealthOverride` were introduced for expert users.
3. A new SBF message `NavCart` was added which combines PVT information from several other SBF blocks.
4. Support for user-defined position datum was added. User needs to provide datum parameters through the commands `setUserDatum`, `setUserDatumVel` & `setUserEllipsoid`.
5. Support for QZSS constellation was added for RTK computation.

2.2 Improvements in version 1.0.0

1. Robustness of logging on the SDCard used in the Evaluation Kit was improved.
2. `ReceiverSetup` SBF block is now output at least once a minute when activated on a port.
3. The command `exeExternalTime` was added to enable OSNMA strict mode.
4. The receiver now uses BeiDou B2a & B1C signals in PVT computation.

2.3 New features in version 1.0.0-beta2

1. Support for some standard position datum (ETRS89, NAD83, etc.) was added.
2. Support for Galileo OSNMA authentication service was added.
3. The receiver now uses BeiDou BDS3 C-NAV signals in PVT computation. Those signals were also default enabled from tracking.
4. Support for DGNSS positioning was added.
5. Support for antenna phase center offset and variations compensation was added.

2.4 Improvements in version 1.0.0-beta2

1. Support for commands `exeEchoMessage` & `setPeriodicEcho` is added.

2. Adding support of ASCIIIn SBF message to record text input messages received on one of the receiver's connection ports.
3. Support for SBF messages ExtEventBaseVectGeod and ExtEventBaseVectCart is added.
4. RTCMv3 messages 1032 and 4091 are now correctly counted as accepted in the InputLink SBF block.
5. The default tracking elevation mask was increased to 5 degrees.
6. GPS L5 signal tracking is now default enabled. Use the command setSignalTracking to control which signals should be tracked by the receiver.

3 Known Issues and Limitations

1. The receiver does not support external SDCard which are formatted with extFAT. FAT32 formatting should be used.
2. The SDCard used in the Evaluation Kit may show disk corruption if logging is ongoing when the power is shut down.
3. Logging SBF or NMEA data on internal disk at a rate of more than 10Hz will result into intermittent data gaps.

4 Support

For further information or support, please consult the Septentrio support website (<http://www.septentrio.com/support>).

5 Legal Notice

Septentrio does not authorize the use of its products as critical components in devices or systems intended for safety-of-life applications or in devices or systems, of which the failure may endanger life or cause injuries, unless written approval is given.

All the firmware and documentation delivered with the mosaic-G5 P3 Firmware Package is licensed, as explained in the About page which is accessible via the web interface of the receiver. On products without web interface, the License can be viewed with the "IstInternalFile, License" command.

6 System Components and Versions

Product: mosaic-G5 P3 Version: 1.0.0 Receiver Platform: GRB0060 Release Date: 31 July 2025	version	mosaic-G5 P3-1.0.0-failsafe.suf	mosaic-G5 P3-1.0.0.suf
Failsafe	2025.04.6-g9fe4076	Y	
Operating System	plp-2025.04.6		Y
GNSS Firmware	2025.04.1-gbfd625f727		Y
Antenna Information	25.1.0		Y